Row-Level Structural Fidelity Diagnostics for Synthetic Tabular Data

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Abstract

Evaluating synthetic tabular data requires auditing individual record integrity in addition to global distribution alignment. Standard statistical metrics (KS complement, Total Variation Distance, Correlation Similarity) assess column distributions in aggregate and are fundamentally blind to localized, row-level conditional dependency violations. We present the Hybrid Integrity Framework (HIF), a row-level structural diagnostic that combines a Logical Sentinel Ensemble (LSE) for categorical conditional dependencies with Neighbor-Invariant Continuity (NIC) for continuous feature consistency. We demonstrate two primary contributions: (1) **Empirically Robust Structural Diagnostics**: across tested generators (Gaussian Copula, Vine Copula, CTGAN, TVAE) and five benchmark datasets (Adult, Credit Default, Census ACS, Online Purchases, Supermarket Sales), HIF identifies row-level dependency ruptures even when aggregate marginal metrics report near-perfect fidelity (KS up to $\approx \mathbf{98\%}$); and (2) **Targeted Remediation**: pruning dependency-violating records ($\boldsymbol{H < 0.5}$) improves downstream predictive utility when structural errors are present (e.g., **+10.4** points F1 on Census ACS CTGAN, paired $\boldsymbol{t}$-test $\boldsymbol{p = 0.002}$; **+12.8** points on Online Purchases CTGAN) while preserving predictive utility in settings where structural errors are weak or absent (Gaussian Copula, paired $\boldsymbol{t}$-test $\boldsymbol{p = 0.13}$). HIF provides practitioners with a practical row-level signal of joint structural integrity for high-stakes synthetic data deployment.

Keywords: Synthetic tabular data; Data integrity; Row-level diagnostics; Dependency auditing; Conditional dependencies; Generative models

1 Introduction

The generation of high-fidelity synthetic tabular data has emerged as a cornerstone of reproducible research in privacy-sensitive domains such as economics [1], finance, and healthcare [2, 3]. With the scaling of deep generative architectures, such as CTGAN [4] and the refinement of flexible statistical models, such as Vine Copulas [5], the community has achieved remarkable success in matching aggregate marginal distributions. However, a critical "Dependency Fidelity Gap" remains: while these models excel at replicating per-column distributions, they frequently produce records that are individually plausible under each column's marginal yet violate the conditional inter-feature dependencies of the real-world joint distribution [6]—violations invisible to any metric that evaluates columns in isolation.

Traditional evaluation pipelines rely on *statistical fidelity* metrics, such as the Kolmogorov-Smirnov test and Total Variation Distance, to assess quality [7, 8]. While useful for verifying marginal distributional alignment, these metrics are inherently *aggregation-blind*: they evaluate each column's distribution independently and cannot detect records that violate conditional inter-feature dependencies yet remain within the expected marginal range of each individual column. Recent attempts to bridge this gap, such as the SHAP-based semantic fidelity metric [9], provide explainability-aware insights but often lack the scalability and technical rigor required for the release of high-stakes synthetic data. As noted by Shang et al. (2026) [10], ensuring the integrity of synthetic tabular data requires more than marginal distributional similarity; it requires a *row-level dependency integrity certificate*. Crucially, this pathology is not exclusive to the opacity of deep neural networks: as we demonstrate, even rigorous statistical models (such as Vine Copulas) confidently hallucinate these joint dependency violations despite strong aggregate marginal fits.

To address these challenges, we introduce the Hybrid Integrity Framework (HIF), a system for detecting logical inconsistencies in synthetic tabular data. HIF reframes the problem of fidelity evaluation as a multidimensional consistency detection task. By moving beyond aggregate statistical snapshots, the HIF identifies logical violations at the record level, enabling the filtering and curation of high-integrity synthetic cohorts. In Section 2, we contextualize our approach within the emerging literature on semantic fidelity and inter-column logic before detailing our methodology (calibration and auditing algorithms in the Appendix).

Our framework comprises three complementary auditing layers:

1. **Logical Sentinel Ensemble (LSE):** An array of discriminative classifiers that identify high-dependency categorical feature hubs in the ground-truth data using cross-validated classification accuracy.
2. **Neighbor-Invariant Continuity (NIC):** A continuous feature auditing mechanism that verifies local semantic adjacency within SVD-embedded categorical contexts using robust percentile thresholding.
3. **Automated Rule Extractor:** An association rule mining engine that extracts high-confidence deterministic implication rules ($A \implies B$) from ground-truth data to enforce hard structural constraints.

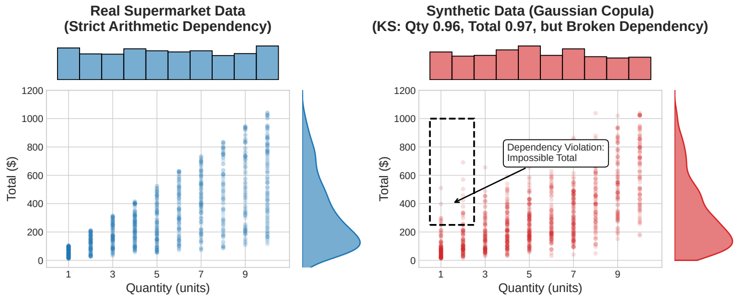

Fig. 1 An empirical demonstration of the Dependency Gap on the Supermarket Sales dataset. While generative models (e.g., Vine Copulas) successfully replicate the 1D marginal distributions of *Quantity* and *Total* (side histograms, KS ≈ 0.96), they fail to preserve the joint arithmetic constraint ($\text{Total} = \text{Price} \times \text{Quantity} \times (1 + \text{Tax})$), hallucinating impossible records (dashed box) that aggregate marginal metrics do not capture.

Summary of Technical Novelty and Core Contributions.

HIF contributes (i) a **Row-Level Structural Diagnostic**: it reframes synthetic data auditing from aggregate column matching to record-level conditional plausibility, combining predictive synergy hub selection with non-compensatory multiplicative aggregation ($\prod_l (1 - \mathcal{P}_l)$) to detect localized structural hallucinations ("impossible junctions") where every individual column value appears marginal-plausible yet their joint combination violates conditional logic; and (ii) **Targeted Remediation**: we characterize when integrity filtering improves downstream utility—on structural-error cohorts (e.g., CTGAN on Census ACS), pruning low-HIF records ($H < 0.5$) yields statistically significant gains (+10.4 points F1, paired t-test $p = 0.002$; +12.8 points on Online Purchases with CTGAN), while on largely error-free cohorts (e.g., Gaussian Copula on Census ACS) filtering preserves utility without detectable degradation (paired t-test $p = 0.13$).

By providing a scalable, data-driven diagnostic, HIF enables the production of synthetic data certified for high-stakes research. The complete open-source implementation, experimental benchmarks, and dataset configurations are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

2 Related Work

Synthetic Tabular Data Generation.

The field has evolved from early copula-based methods to deep generative architectures. GAN-based models such as CTGAN and TVAE [4] set an early benchmark for capturing joint distributions by building on the Synthetic Data Vault framework [11], later improved for skewed distributions by CTAB-GAN [12]. Diffusion paradigms such as TabDDPM [13], flow-based gradient-boosted tree generators [14], and composable architectures such as CoDi [15] have pushed the frontier of manifold matching; permutation-invariant objectives [16] and federated frameworks [17] extend synthesis to column-order robustness and decentralized settings. However, both statistical and deep architectures suffer a fundamental objective mismatch: statistical models (such as Vine Copulas) excel at per-column marginal fitting but smooth over complex non-linear conditional constraints, while neural models (such as CTGAN) can learn non-linear categorical contexts but remain prone to mode collapse or localized joint violations. Our work focuses on a representative spectrum of generators—from classical *Gaussian Copulas* [18] and *Vine Copulas* [5] to neural architectures such as *CTGAN*—and shows that the Dependency Gap recurs across these diverse generative paradigms in our tested settings.

Evaluation Metrics and the Dependency Gap.

Traditional evaluation pipelines primarily rely on statistical fidelity (e.g., Kolmogorov-Smirnov tests, Correlation Distance) and predictive utility (Train-on-Synthetic, Test-on-Real; TSTR). Although effective for global assessment, these metrics evaluate aggregate column-level statistics and are therefore blind to row-level conditional dependency violations—by design, a corruption that samples each feature from its correct marginal but ignores inter-feature correlations is entirely invisible to such metrics. Long et al. (2025) [6] were among the first to formalize these "Dependency Gaps," proposing rule-based audits for inter-column relationships. Umesh et al. [19] studied the preservation of logical and functional dependencies from a generation-side perspective, constraining the generative model to respect mined dependencies at training time; this is complementary to—but distinct from—a post-hoc audit of an arbitrary generator's output, which is the setting we address, and dependency-aware representations have recently been shown to enhance tabular understanding [20]. Similarly, Yu et al. (2025) [9] introduced the SHAP Distance, an explainability-aware fidelity metric, but it yields a dataset-level attribution summary rather than per-record joint-conditional scores and scales exponentially with the feature count (see Appendix D.2). Notably, distribution-level auditing faces provable limits: identity and uniformity tests require sample sizes that scale with the domain support [21–23], a cost only partially mitigated even for product-structured distributions [24], and conditional independence testing is likewise sample-intensive in high dimensions [25]. HIF automates dependency discovery through predictive synergy scoring: unlike rule-mined audits, which are binary and require manual specification, HIF provides a soft probabilistic integrity score using non-compensatory multiplicative aggregation that prevents high-fidelity feature dimensions from masking localized dependency violations.

3 Mathematical Foundation

The Hybrid Integrity Framework (HIF) formalizes synthetic data auditing as a probabilistic conditional scoring task. Let $\mathcal{D}_{real}$ denote the original training dataset. We define the **ground-truth manifold** $\mathcal{M}_{real}$ as the support of the joint distribution $P_{real}(\mathbf{x})$—the localized subspace of valid feature configurations that occur in reality [26]. A synthetic record $\mathbf{x} = [\mathbf{x}_{cat}, \mathbf{x}_{cont}]$ (with $\mathbf{x}_{cat}$ and $\mathbf{x}_{cont}$ the sub-vectors of categorical and continuous features) is a *dependency-violating record* if it is plausible under each marginal distribution yet falls outside the ground-truth manifold (i.e., it is highly improbable under the joint conditional distribution $P_{\mathcal{D}_{real}}(\mathbf{x}_h \mid \mathbf{x}_{\setminus h})$ learned from the data). This distinction is central: aggregate fidelity metrics (KS, TVD, JCD)

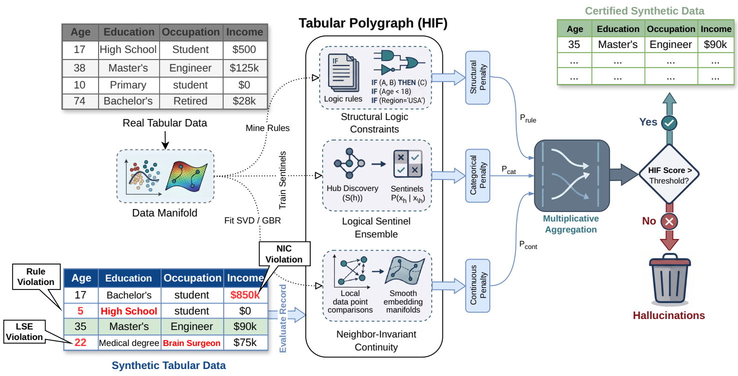

Fig. 2 Architecture of the Hybrid Integrity Framework (HIF). Synthetic tabular data is verified by the Logical Sentinel Ensemble (LSE) and Neighbor-Invariant Continuity (NIC) modules, combined via multiplicative aggregation to filter semantically inconsistent records.

assess whether records fall within the expected marginal range of each column in isolation, whereas HIF additionally evaluates whether each record's joint configuration structurally aligns with the ground-truth manifold. The complete calibration process for establishing these data-driven manifold boundaries is detailed in Algorithm 1.

3.1 Logical Sentinel Ensemble (LSE)

To audit categorical consistency, we first identify the set of Dependency Hubs $\mathcal{H}$ that govern the structural dependencies of the dataset. For a given row $\mathbf{x}$, let $x_h \in \mathcal{X}_h$ denote the categorical value of a target feature h, and let $\mathbf{x}_{\setminus h}$ denote the values of all other features (i.e., $\mathbf{x}$ with feature h removed).

We train an independent Random Forest classifier [27]—leveraging decision tree recursive partitioning to capture non-linear feature interactions [28]—which we call a *Sentinel* and denote as f_h, to predict x_h given $\mathbf{x}_{\setminus h}$. The sentinel outputs a conditional probability distribution $P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h})$ for all possible categories $c \in \mathcal{X}_h$. We define the importance of a feature h using the Feature Dependency Score $S(h)$, the cross-validated classification accuracy of its sentinel:

$$S(h) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{real}} \left[\mathbb{I} \left(\arg\max_{c \in \mathcal{X}_h} P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h}) = x_h \right) \right] \tag{1}$$

where $\mathbb{I}$ is the indicator function. By maximizing $S(h)$, we select hubs that are most strongly constrained by conditional interactions with other attributes. While computationally efficient, this reliance on single-variable classification accuracy primarily captures univariate dependencies and may under-represent higher-order parity relationships; future work will investigate Total Correlation and multi-way interaction kernels to broaden hub selection.

Consider a demographic dataset where x_h is *Education Level* and $\mathbf{x}_{\setminus h}$ contains *Age* (e.g., 14). Because a 14-year-old cannot hold a PhD in the real world, the sentinel f_h learns from the ground-truth data to assign $P_{f_h}(\text{PhD} \mid \text{Age} = 14) \approx 0$. If a generative model hallucinates a 14-year-old with a PhD, the sentinel will flag the record because the assigned probability for the hallucinated category is extremely low.

Confidence Floor Calibration.

Drawing inspiration from distribution-free quantile calibration in inductive conformal prediction [29–31], we establish a data-driven Confidence Floor δ_h for each hub based on empirical calibration error bounds, defined as the empirical 5th-percentile ($Q_{0.05}$) of the *out-of-bag* probability mass assigned to the correct category by the sentinel in the ground-truth data, with a safety minimum of 0.01:

$$\delta_h = \max\left(Q_{0.05}\left(\{P_{f_h}(x_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}\right), 0.01\right) \tag{2}$$

Using out-of-bag predictions (i.e., predictions of the trees that did not train on each row) yields out-of-sample floors without requiring a held-out split, preventing over-optimistic calibration on rows the sentinels memorized during training. This 5th-percentile floor acts as a conservative safety margin: if the real data occasionally contains 18-year-olds with college degrees, the sentinel's probability might be low but non-zero (e.g., 0.05), and the floor δ_h ensures we do not penalize rare but valid records that fall within the natural density of the ground-truth manifold.

The per-hub penalty $\mathcal{P}_h(\mathbf{x})$ uses a soft-threshold activation to suppress noise-level deviations, harshly penalizing only those records whose probabilities fall far below the confidence floor (such as the impossible 14-year-old PhD):

$$\mathcal{P}_h(\mathbf{x}) = \text{clip}\left(\frac{\delta_h - P_{f_h}(x_h \mid \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right) \tag{3}$$

The categorical penalty aggregates multiplicatively across hubs, ensuring that ruptures in multiple manifold dimensions are compounded.

$$\mathcal{P}_{cat}(\mathbf{x}) = 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h(\mathbf{x})) \tag{4}$$

The complete manifold calibration procedure—hub selection, confidence-floor calibration, NIC regressor training, and rule mining—is summarized in Algorithm 1 (Appendix B).

3.2 Neighbor-Invariant Continuity (NIC)

For continuous features $\mathbf{y}$, we evaluate integrity via Semantic Adjacency. We map the categorical context to a continuous latent space $\mathcal{Z}$ using a standardized spectral projection $\phi : \mathcal{X}_{cat} \to \mathcal{Z}$—drawing on classical Multiple Correspondence Analysis (MCA) and Singular Value Decomposition (SVD) on binary indicator matrices [32, 33]. Standardizing the indicator matrices by their standard deviation prior to SVD mathematically replicates the inverse-frequency weighting of MCA, ensuring that rare but highly informative categorical constraints dominate the continuous manifold projection. We then model the conditional expectation of each continuous feature y as a non-linear function of this manifold using a Gradient Boosting Regressor r_y [34, 35]:

$$\hat{y}(\mathbf{x}) = \mathbb{E}[y \mid \phi(\mathbf{x}_{cat})] \approx r_y(\phi(\mathbf{x}_{cat})) \tag{5}$$

For any given record $\mathbf{x}$, let $e_y(\mathbf{x}) = |y(\mathbf{x}) - \hat{y}(\mathbf{x})|$ denote the absolute regression residual, and define the ground-truth residual set as $\mathcal{E}_y = \{e_y(\mathbf{x}) \mid \mathbf{x} \in \mathcal{D}_{real}\}$. For example, a synthetic 45-year-old Senior Executive generated with an Income of \$12 yields an enormous residual against the expected executive income ($\hat{y} \approx$ \$120,000), far exceeding the natural variance of executives in the real dataset and signaling a continuous dependency gap. To handle extreme feature skewness (common in census domains), we define a robust hybrid threshold τ_y from $\mathcal{E}_y$ using the Median Absolute Deviation (MAD) [36], a robust estimate of local manifold spread that is less sensitive to outliers than the standard deviation:

$$\tau_y = \max\left(Q_{0.95}(\mathcal{E}_y), \mathrm{Median}(\mathcal{E}_y) + 2 \cdot \mathrm{MAD}(\mathcal{E}_y)\right) \tag{6}$$

This threshold keeps the NIC layer stable even in high-skew distributions, such as macroeconomic wage data.

The continuous penalty $\mathcal{P}_{cont}(\mathbf{x})$ uses a dynamic scaling factor $\gamma_y = \max(Q_{0.98}(\mathcal{E}_y), 2\tau_y, 3\mathrm{MAD}(\mathcal{E}_y), 0.1)$ that bounds the penalty relative to the feature's natural noise floor, so small deviations beyond τ_y yield soft penalties while massive violations (like the \$12 executive) yield a maximum penalty of 1.0:

$$\mathcal{P}_{cont}(\mathbf{x}) = \max_y \left[\mathrm{clip}\left(\frac{e_y(\mathbf{x}) - \tau_y}{\gamma_y}, 0, 1\right)\right] \tag{7}$$

As a defensive guard, NIC detects *collapsed* continuous features whose conditional predictions have near-zero variance (i.e., the feature is effectively constant given the categorical context); for such degenerate features, all records receive a uniform base penalty since the expected residual distribution provides no reliable baseline. This guard did not activate on any benchmark dataset in our experiments.

3.3 Multiplicative Aggregation

In contrast to additive fidelity metrics that allow a model to "compensate" for logical failures with high statistical overlap, HIF formalizes integrity as an Algebraic Intersection. Drawing from the structured drift-proof framework [10], the hybrid integrity score $H(\mathbf{x})$ is defined as the geometric mean of the component validities:

$$H(\mathbf{x}) = \left(\prod_{l \in \{cat, cont, rule\}} (1 - \mathcal{P}_l(\mathbf{x}))\right)^{1/L} \tag{8}$$

where l iterates over the active auditing layers. The geometric mean preserves the non-compensatory property of a Logical AND gate: a significant failure in any single manifold dimension ($\mathcal{P}_l \to 1$) drives the entire record integrity toward zero, regardless of how well the other components align (see Table 4, Section 4.5). For a formal analysis of this aggregation logic under explicit assumptions—intersection soundness (Theorem 1), asymptotic convergence (Theorem 2), and the Type I false-rejection bound under quantile calibration (Theorem 3)—we direct the reader to **Appendix A**.

While HIF prioritizes soft probabilistic validities via LSE and NIC, we incorporate symbolic implication rules as an optional *Structural Logic Constraint* layer ($l = rule$) to enforce known deterministic or physical laws (e.g., Age ≥ 18 for voters), bridging differentiable manifolds and rigid symbolic constraints.

The Integrity Frontier.

The dataset-level HIF score is the expected record-level integrity. To provide a discrete diagnostic, a record is classified as a *Semantic Violation* if $H(\mathbf{x}) < 0.5$, defining the framework's Integrity Frontier.

$$\text{HIF Score} = \frac{1}{N} \sum_{i=1}^{N} H(\mathbf{x}_i) \tag{9}$$

This score provides a bounded diagnostic $\in [0, 1]$, where 1.0 indicates a logically certified synthetic cohort. The complete end-to-end auditing procedure for scoring a synthetic dataset is detailed in Algorithm 2 (Appendix B).

Because HIF penalizes records that violate *learned conditional dependencies* rather than marginal rarity—the confidence floor δ_h is calibrated on the ground-truth manifold precisely to avoid penalizing rare-but-valid tail records—filtering operates on structural consistency rather than demographic support; a systematic audit of representation drift across sensitive attributes under integrity filtering is left to future work. A primary critique of multiplicative evaluation is that absolute scores often appear lower than the near-unity scores (≥ 0.95) typical of aggregate 1D distributional metrics in standard evaluation benchmarks [2, 6, 11]. This is a deliberate design property: a record with a severe violation of any single dependency should not receive a high integrity score, regardless of how well its remaining features conform to marginal expectations. Assuming conditional independence of error dimensions given the feature representations, the product of validities $\prod(1 - \mathcal{P}_l)$ serves as a pseudo-likelihood proxy for the joint structural conformity of the record, and under empirical quantile calibration ($\delta_h = Q_\alpha$, $\gamma_y = Q_{1-\beta}$) the false-rejection probability for valid tail records is upper-bounded by the Union Bound under stated assumptions (Theorem 3, Appendix A).

4 Results

We evaluated the Hybrid Integrity Framework (HIF) on five benchmark datasets with diverse structural complexity: U.S. Census ACS [37, 38], Adult Income, Credit Card

Default, Online Purchases, and Supermarket Sales. Experiments used $N = 3$–10 random seeds depending on the evaluation; we report mean $\pm$ standard deviation (SD) unless otherwise noted.

To calibrate the framework's response, we executed a controlled semantic violation protocol: we stochastically permuted (shuffled) the values of specific features across different rows within highly constrained categorical-continuous dependencies for a controlled fraction of records $\eta \in [0, 0.6]$ (Census ACS, $N = 3$ seeds). Shuffling existing values across rows rather than injecting external noise preserves each column's marginal distribution (satisfying standard MM and KS metrics) while breaking the underlying joint conditional structure at the row level. Under this protocol, the HIF score exhibited strong monotonicity in our runs (Spearman $\rho = -1.0$ across seeds): the mean HIF decreased from 0.941 ± 0.003 at $\eta = 0$ to 0.712 ± 0.002 at $\eta = 0.6$, whereas the marginal metrics—Moment Matching (MM) and Kolmogorov-Smirnov (KS)—remained effectively static ($\rho \approx 0$), since the shuffled values are drawn from the intact marginals. Notably, the Joint Correlation Distance (JCD) also decayed sharply ($88.8 \rightarrow 65.0$, $\rho = -1.0$): shuffling a feature destroys its pairwise correlations, so JCD is a sensitive aggregate drift detector—but it yields only a dataset-level summary and cannot attribute degradation to specific records. This motivates HIF's row-level output and the low-rate, marginals-preserving conditional-swap protocol used in Section 4.4.

Evaluation Protocol and Threats to Validity.

We distinguish ranking quality (ROC-AUC, PR-AUC) from thresholded classification quality (F1), treating the former as threshold-independent diagnostics of error ranking and the latter as an operating-point metric. For baseline comparability, IF and LOF use a `contamination` parameter set to the known corruption rate in synthetic stress tests; this favors those baselines for F1 and is reported explicitly to avoid hidden calibration advantages. The controlled corruption protocol above is used strictly for metric response calibration, not as the primary benchmark for diagnostic utility: our main empirical validation assesses HIF across three independent regimes—(1) a cross-architecture maturity audit of generative models (Section 4.1), (2) downstream predictive utility on un-manipulated synthetic cohorts (Section 4.2), and (3) held-out error families that HIF was not specifically engineered to detect (plus the targeted dependency-violation family as a positive control; Section 4.3). Sample sizes differ by experiment: the significance, held-out, and cross-architecture audits use $N = 10$ seeds with paired tests and exact p-values where applicable, whereas the calibration-domain experiments use $N = 3$ seeds and are reported as descriptive mean $\pm$ SD without inferential claims; the calibration-domain and statistical-significance analyses are confined to Census ACS, so their inferential conclusions should not be extrapolated to other domains without further evaluation. Residual threats include dataset selection bias and the possibility that rare-but-valid tail patterns are partially coupled with high-penalty regions.

4.1 Cross-Architecture Maturity Audit and Cohort Certification

To evaluate the diagnostic utility of HIF across generative paradigms, we conducted a cross-architecture audit across four benchmark datasets from diverse domains (Table 1): Supermarket Sales (retail arithmetic constraints), Online Purchases (e-commerce transaction constraints), Credit Card Default (financial classification), and Adult Census Income (demographic benchmark). The results reveal a key insight:

1. Integrity-Fidelity Decoupling on Constrained Domains.

Across all four datasets, Vine Copulas achieve high marginal statistical fit (KS $\geq$ 0.969). However, on datasets with cross-feature relational constraints (Supermarket Sales and Online Purchases), Vine Copulas exhibit severe joint dependency failures, with violation rates of **86.2% to 99.9%** (HIF $\leq$ 0.271); CTGAN and TVAE show similar deficits on transaction constraints (HIF = 0.02–0.28, violation rates 84.9%–100%). Marginal metrics (KS) report these models as high-fidelity, underscoring that aggregate fidelity metrics do not capture joint dependency ruptures. A qualitative audit on Supermarket Sales (Vine Copula) confirmed the mechanism: generators match each column's marginal distribution yet violate the arithmetic identity subtotal = unit_price $\times$ quantity $\times$ $(1 + \text{tax})$, with HIF flagging essentially the entire cohort (violation rate 99.9% across $N = 10$ seeds; component rates 100.0% LSE, 63.2% rules, 0.0% NIC)—while KS and joint-correlation metrics reported high fidelity.

2. Evaluation on Unconstrained Manifolds.

On unconstrained demographic and financial benchmarks (Adult Income, Credit Default), raw generators preserve conditional structures better, with HIF scores ranging from 0.33 to 0.98 (violation rates 0.5%–67.6%), providing a continuous gradient of structural integrity for comparing generator maturity. Notably, TVAE achieves the highest HIF on both Credit Default (0.827) and Adult Income (0.981), demonstrating that VAE-based architectures can excel at preserving joint dependencies on unconstrained domains.

3. Quantitative Decoupling Evidence.

Pearson correlation analysis across all 16 dataset–generator configurations of the four audit benchmarks (Census ACS is held out as the calibration benchmark; Table 1) characterizes the decoupling precisely: HIF shows no linear relationship with marginal fidelity ($\rho(\text{HIF}, \text{KS}) = 0.038$) and only a weak relationship with support coverage ($\rho(\text{HIF}, \alpha) = 0.325$), but a strong positive correlation with joint correlation fidelity ($\rho(\text{HIF}, \text{JCD}) = +0.82$). HIF thus aligns with joint-dependency structure while being decoupled from per-column marginal metrics: high marginal fidelity (KS) does not imply high integrity, exactly the pattern observed for Vine Copulas on constrained domains.

Table 1 Cross-Architecture Audit: Statistical Fidelity (KS), Support Overlap (α-Precision, β-Recall), and Joint Dependency Integrity (HIF) across benchmark datasets ($N = 10$ seeds, Mean $\pm$ SD). High marginal fidelity (KS > 0.96) or high support coverage (α-Prec > 0.79) does not imply joint conditional integrity. On datasets with cross-feature constraints, models exhibit up to 100% dependency violation rates.

Generator	KS (↑)	α-Prec (↑)	β-Rec (↑)	HIF (↑)	Viol. Rate (↓)
Supermarket Sales					
Gaussian Copula	0.903 ± 0.006	0.892 ± 0.012	0.918 ± 0.013	**0.963 ± 0.006**	**2.6 ± 0.6%**
Vine Copula	**0.975 ± 0.003**	0.792 ± 0.011	0.862 ± 0.016	0.025 ± 0.002	99.9 ± 0.1%
CTGAN	0.771 ± 0.029	0.644 ± 0.071	0.861 ± 0.027	0.019 ± 0.010	100.0 ± 0.0%
TVAE	0.881 ± 0.011	0.827 ± 0.038	0.953 ± 0.015	0.021 ± 0.001	100.0 ± 0.0%
Online Purchases					
Gaussian Copula	0.753 ± 0.010	0.489 ± 0.013	0.819 ± 0.023	**0.626 ± 0.009**	**40.3 ± 1.3%**
Vine Copula	**0.969 ± 0.004**	0.831 ± 0.019	0.942 ± 0.013	0.271 ± 0.008	86.2 ± 1.3%
CTGAN	0.677 ± 0.068	0.524 ± 0.061	0.882 ± 0.047	0.278 ± 0.047	84.9 ± 5.7%
TVAE	0.854 ± 0.019	0.754 ± 0.049	0.818 ± 0.035	0.119 ± 0.008	99.0 ± 0.5%
Credit Default					
Gaussian Copula	0.747 ± 0.002	0.729 ± 0.009	0.951 ± 0.010	0.612 ± 0.005	30.5 ± 0.8%
Vine Copula	**0.981 ± 0.001**	0.801 ± 0.011	0.813 ± 0.014	0.401 ± 0.007	55.7 ± 1.1%
CTGAN	0.717 ± 0.059	0.627 ± 0.066	0.859 ± 0.018	0.333 ± 0.038	67.6 ± 4.8%
TVAE	0.878 ± 0.005	0.961 ± 0.009	0.935 ± 0.007	**0.827 ± 0.009**	**13.5 ± 1.1%**
Adult Income					
Gaussian Copula	0.848 ± 0.006	0.939 ± 0.006	0.838 ± 0.016	0.830 ± 0.007	10.5 ± 0.8%
Vine Copula	**0.981 ± 0.005**	**0.964 ± 0.008**	0.799 ± 0.021	0.764 ± 0.004	15.9 ± 0.9%
CTGAN	0.682 ± 0.086	0.794 ± 0.058	0.802 ± 0.029	0.659 ± 0.013	29.4 ± 1.8%
TVAE	0.904 ± 0.019	0.701 ± 0.036	0.756 ± 0.017	**0.981 ± 0.002**	**0.5 ± 0.2%**

4.2 Alignment with Downstream Utility

A critical requirement for any synthetic data metric is its ability to act as a proxy for downstream predictive utility, a validation paradigm now standard in privacy-sensitive domains [39]. We measured downstream TSTR F1 under integrity filtering across representative generative architectures (Table 2), using a Random Forest classifier [27] (via *Scikit-learn* [40]); the *Rule-only Baseline* prunes records violating any mined implication rules, whereas the *HIF Filtered* variant uses the semantic frontier ($H < 0.5$).

The paired tests on Census ACS ($N = 10$ seeds, binary median-split `household_income` target) quantify the asymmetry. For the Gaussian Copula, filtering yields $p = 0.133$ (95% CI $[-0.008, +0.001]$): no detectable utility degradation, as expected for a cohort largely free of conditional dependency violations (violation rate $< 1\%$). For CTGAN, downstream utility gains are statistically significant (paired t-test $p = 0.002$; Wilcoxon $p = 0.002$; 95% CI $[+0.049, +0.160]$; paired Cohen's $d \approx 1.34$), with F1 improving from 0.427 ± 0.090 to 0.532 ± 0.139 ($+0.104$), supporting the claim that HIF is particularly useful for architectures that produce joint dependency violations. On Online Purchases with the Gaussian Copula (40% violation rate), filtering retains 59.7% of records and F1 moves within noise ($0.977 \to 0.968$), tempering the hypothesis that dependency violations always act as predictive noise: whether removing them improves utility depends on the strength of the violation-to-label association.

Table 2 Downstream utility filtering. Selecting records that satisfy the manifold laws maintains or improves predictive performance while pruning logically inconsistent records. All rows use $N = 10$ seeds.

Dataset	Generator	Variant	Retention (%)	F1 ($\pm$ SD)	Accuracy ($\pm$ SD)
ACS (Census)	Gaussian Copula	Full synthetic	100.0	0.940 $\pm$ 0.006	0.940 $\pm$ 0.006
	Gaussian Copula	**HIF Filtered**	**99.2**	0.937 $\pm$ 0.006	0.937 $\pm$ 0.006
	CTGAN	Full synthetic	100.0	0.427 $\pm$ 0.090	0.518 $\pm$ 0.058
	CTGAN	**HIF Filtered**	**65.7**	**0.532** $\pm$ 0.139	**0.601** $\pm$ 0.090
Purchases	Gaussian Copula	Full synthetic	100.0	0.977 $\pm$ 0.010	–
	Gaussian Copula	**HIF Filtered**	**59.7**	0.968 $\pm$ 0.024	–

To characterize when integrity filtering recovers utility across domains, we tested the paired F1 difference (HIF-filtered minus full synthetic) across all dataset–generator combinations of the cross-architecture benchmark with $N = 10$ seeds. Significant utility recovery occurs in four configurations: Census ACS (ΔF1 $= +0.104$, $p = 0.002$ for CTGAN; $+0.195$, $p < 0.001$ for Vine Copula) and Online Purchases (ΔF1 $= +0.128$, $p = 0.037$ for CTGAN; $+0.399$, $p < 0.001$ for Vine Copula). Recovery is selective rather than a deterministic consequence of the violation rate: high-violation cohorts on Credit Default show no significant change despite 67.6% and 55.7% violation rates (CTGAN, ΔF1 $= +0.008$, $p = 0.29$; Vine Copula, -0.002, $p = 0.16$), and Online Purchases Gaussian Copula shows none at 40.3% (-0.010, $p = 0.28$). Whether removing flagged records improves utility therefore depends on the strength of the association between the violated conditional structure and the downstream target, not on the violation rate alone. Conversely, most remaining configurations exhibit small, non-significant changes (e.g., Census ACS Gaussian Copula, ΔF1 $= -0.003$, $p = 0.13$; Adult TVAE, -0.002, $p = 0.48$; Adult Gaussian Copula, -0.007, $p = 0.22$), confirming that filtering does not penalize well-formed synthetic data. Only one configuration shows a small but significant degradation (Credit Gaussian Copula, ΔF1 $= -0.021$, $p = 0.013$), indicating that pruning can also discard flagged records that retain predictive signal for the target; this remains the exception rather than the rule. Finally, four transaction configurations (Supermarket Sales CTGAN/TVAE/Vine; Online Purchases TVAE) retain at most $\approx 2\%$ of records, leaving the retained cohort too small to estimate utility reliably, and their F1 deltas are consequently not reported. Together these results indicate that HIF-driven filtering is a targeted remediation: it recovers utility where structural errors degrade the target signal and is otherwise utility-neutral.

4.3 Comparison with Outlier Detection Baselines

A natural question is whether standard outlier detection methods—Isolation Forest (IF) and Local Outlier Factor (LOF)—can serve as alternatives to HIF. Table 3 compares detection performance across five error types on Census ACS at 40% corruption ($N = 10$ seeds). LOF and IF are configured with a `contamination` parameter set to the known corruption rate (0.4), giving them a tuned decision threshold for F1 in this stress-test setup, whereas HIF uses a fixed non-parametric default threshold ($H > 0.5$).

Under threshold-independent ROC-AUC, HIF strongly outperforms baseline detectors on *Conditional Dependency Violations* (ROC-AUC = 0.830, PR-AUC = 0.789), whereas Isolation Forest is near-random on marginal-preserving dependency swaps (ROC-AUC = 0.512) and LOF achieves lower ranking quality (ROC-AUC = 0.719, PR-AUC = 0.654). Conversely, generic outlier detectors excel at catching broad geometric feature outliers (e.g., random noise injection), where LOF reaches ROC-AUC = 0.864; HIF also maintains high sensitivity on random noise (ROC-AUC = 0.890) because breaking marginals simultaneously disrupts conditional dependencies. On distribution-level shifts (covariate shift), Isolation Forest outperforms HIF (ROC-AUC 0.490 vs. 0.289), indicating that HIF is a specialized diagnostic for row-level joint conditional integrity rather than global distribution drift.

Why do standard outlier detectors fail on Dependency Gaps? Isolation Forest and LOF evaluate geometric distance or marginal density across the entire continuous feature space. If a synthetic record hallucinates an impossible logic combination (e.g., a 14-year-old with a PhD) where both values are common marginally, the record resides in a dense region of the globally projected feature space; because Isolation Forest splits features independently, it rarely isolates the specific interacting dimensions that violate the conditional rule. A Dependency Gap is a structural, semantic error, not a geometric density anomaly—which is precisely why HIF's logic-aware sentinels succeed where geometric detectors fail.

Table 3 Held-Out Error Detection: HIF vs. Outlier Baselines (Census ACS, 40% corruption). We report threshold-independent ROC-AUC and F1 score across 10 random seeds. HIF achieves strong discrimination on conditional dependency violations (ROC-AUC = 0.830) where Isolation Forest is near-random (ROC-AUC = 0.512). Geometric detectors (LOF, IF) excel at broad marginal noise where HIF is intentionally permissive.

	ROC-AUC (↑)			F1 Score (↑)		
Error Type	**HIF**	**IF**	**LOF**	**HIF**	**IF**	**LOF**
Random Noise Injection	**0.890**	0.521	0.864	0.369	0.428	**0.617**
Row Duplication	0.506	0.501	**0.508**	0.018	0.404	**0.535**
Feature Dropout	**0.652**	0.280	0.489	0.072	0.157	**0.518**
Covariate Shift	0.289	**0.490**	0.211	0.007	**0.398**	0.289
Conditional Dependency Violations	**0.830**	0.512	0.719	0.224	0.413	**0.593**

4.4 Differential Diagnostic: Integrity vs. Loyalty

To determine whether HIF provides unique utility over joint statistical metrics, we evaluated its response at low corruption levels ($p \in [0.001, 0.1]$) using a conditional-swap protocol that preserves marginal distributions (Census ACS, $N = 3$ seeds). At these low rates, the HIF score decays monotonically with p (Spearman $\rho = -1.0$; from 0.941 at $p = 0$ to 0.919 at $p = 0.1$), whereas the aggregate marginal metrics (Moment Matching, KS) remain static and the Joint Correlation Distance moves only marginally ($88.8 \rightarrow 87.9$, relative shift $< 1.2\%$). The small absolute JCD change reflects dilution:

a few thousand corrupted rows contribute negligibly to a cohort-averaged correlation statistic. Because HIF emits a per-record penalty, it localizes exactly which rows carry the corrupted conditional relationships—a capability no aggregate metric provides. This reading is complementary, not adversarial: under the broad, high-rate permutation protocol described earlier (which destroys pairwise correlations), the JCD decays sharply in lockstep with the HIF score. The two tools therefore address related but distinct failure modes: the JCD is a sensitive dataset-level drift detector, while HIF provides record-level attribution of conditional dependency violations.

4.5 Component Ablation Study

To understand the contribution of each HIF component, we performed an ablation study on Census ACS using CTGAN ($N = 5$ seeds; Table 4). Because Census ACS is dominated by complex, non-linear categorical dependencies, the Logical Sentinel Ensemble (LSE) is the most critical component, raising F1 from 0.422 ± 0.044 (no filtering) to 0.719 ± 0.086 while retaining 27.7% of records. In contrast, rule-based filtering removes few records (2.2% violation rate) and leaves utility near baseline, and NIC-only pruning barely changes retention (13.5% violation rate), since Census ACS numeric features carry few categorical-driven conditional failures; combining LSE with NIC yields intermediate behavior (retention 41.9%, F1 0.639 ± 0.089).

The ablation also illustrates the aggregation method's role: Full HIF with multiplicative product aggregation ($F1 = 0.513 \pm 0.075$) outperforms the arithmetic mean ($F1 = 0.432 \pm 0.047$). The multiplicative product acts as a non-compensatory (Logical AND) gate—preventing a high score in one dimension from hiding a severe failure in another—pruning an additional 30% of records relative to arithmetic aggregation (retaining 64.9% vs. 95.2%). The magnitude of this advantage should be read with caution: retention and utility are confounded, and the gap is driven primarily by how aggressively each variant prunes. Consistent with this interpretation, the aggregation choice is immaterial in settings free of structural errors: under Gaussian Copula on Census ACS (violation rate $< 1\%$), all ablation configurations yield statistically indistinguishable F1 (0.934–0.941).

Understanding the Violation Rate Drop.

It may seem counterintuitive that adding components (NIC, Rules) to the Logical Sentinel Ensemble (LSE) decreases the violation rate and lowers F1 compared to LSE alone. This behavior stems from aggregating continuous probabilities rather than using boolean rejection. When LSE assigns a harshly low score to a synthetic record, LSE-only will flag it as a violation (leading to a high rejection rate and high F1 on the small remaining cohort). However, if the record is structurally sound in continuous features and rules, NIC and Rule modules will assign it scores near 1.0. Averaging these continuous scores pulls the overall HIF score up, often rescuing the borderline record from falling below the $H < 0.5$ threshold. Consequently, Full HIF acts as a balanced filter that retains a larger, more diverse dataset (yielding an F1 of 0.513 with 64.9% retention) rather than an ultra-strict, low-diversity filter (LSE-only F1 0.719 with 27.7% retention). The geometric mean is used specifically because it is harder to "rescue" a low score via multiplication than arithmetic addition, pruning more aggressively while avoiding total cohort collapse.

Table 4 Component Ablation Study on Census ACS (CTGAN, $N = 5$ seeds). The multiplicative product implements a non-compensatory Logical AND gate, pruning more records than the arithmetic mean and reaching the highest F1 on a cohort with structural errors. In cohorts free of structural errors (Gaussian Copula), all configurations are statistically indistinguishable (F1 0.934–0.941).

Ablation	**Retention (%)**	**F1 (mean ± SEM)**	**Violation Rate**
No filtering	100.0%	0.422 ± 0.044	0.0%
LSE-only	27.7%	0.719 ± 0.086	72.3%
NIC-only	86.6%	0.418 ± 0.044	13.5%
Rules-only	97.8%	0.424 ± 0.041	2.2%
LSE + NIC	41.9%	0.639 ± 0.089	58.1%
Full HIF (arith.)	95.2%	0.432 ± 0.047	4.8%
Full HIF (multiplicative)	**64.9%**	**0.513 ± 0.075**	**35.1%**

5 Discussion

The results indicate that the Hybrid Integrity Framework provides a conservative diagnosis in our experimental settings. By employing a multiplicative aggregation model, HIF ensures that a failure in any single semantic dimension results in a low integrity score. This non-compensatory evaluation is important for high-stakes domains, where a single inconsistent record can invalidate an analysis.

5.1 Reconciling Generative Paradigms: Statistical vs. Neural Trade-offs

A central finding of our cross-architecture audit (Table 1) is that the Dependency Gap is not restricted to deep neural black-boxes; it appears as a recurring pathology affecting both statistical and neural generative models in distinct ways. In our runs, statistical copulas preserve integrity well on Census ACS and Supermarket Sales (HIF 0.94–0.96 for the Gaussian Copula) but degrade on Online Purchases and Credit Default (HIF 0.61–0.63), and Vine Copulas—despite the highest marginal fit (KS $\geq$ 0.969)—catastrophically violate transaction arithmetic constraints (HIF 0.02–0.27). Conversely, neural models such as CTGAN capture discrete categorical clusters on demographic data (HIF 0.33–0.66) but struggle with continuous arithmetic identities (HIF 0.02 on Supermarket Sales). This reconciles our core motivation: the Dependency Gap does not stem from neural opacity alone, nor from copula rigidity. It is an inherent artifact of standard generative loss functions, which optimize aggregate distributional fit (likelihood, Wasserstein distance, or marginal discrimination) rather than row-level joint conditional integrity—so post-hoc row-level auditing is necessary regardless of whether the underlying generator is statistical or neural.

5.2 Practical Interpretation, Specificity, and Scalability

Because HIF computes the geometric mean of the component validities across L auditing layers, the overall score $H(\mathbf{x}) = (\prod_l \mathcal{C}_l)^{1/L}$ requires simultaneous joint satisfaction of all conditional models—an intentional conservative design for high-stakes auditing. This specificity–sensitivity trade-off is deliberate: HIF prioritizes *dependency specificity*, flagging violations of learned conditional relationships rather than marginal distributional jitter. A generic anomaly detector sensitive to marginal outliers (such as LOF) would produce false positives on valid but rare records; HIF's sensitivity is bounded by the confidence floors learned from the real data distribution.

The LSE architecture is highly scalable, with a computational complexity of $O(N \cdot K)$ for auditing N records across K hubs, and scales effectively to tables with thousands of columns. Our empirical sweep (Appendix D.1) shows stable performance even at $K = 1$, meaning the hub count can be minimized to reduce the K-multiplier overhead. In our benchmarks, auditing 10,000 synthetic rows required $\approx 1.5\,\mathrm{s}$ on standard multi-core commodity hardware. However, the memory overhead of training and storing K Random Forest sentinels for ultra-high-dimensional tables (e.g., $D > 10,000$) remains a potential hardware bottleneck; in such regimes, sparse-linear or neuro-inspired models offer a better trade-off between semantic depth and memory usage.

5.3 Limitations and Failure Regimes

Despite strong results, several limitations remain. First, HIF is not designed as a global covariate-shift detector and can underperform methods such as Isolation Forest on broad geometric drift. Second, the quality of hub selection and confidence-floor calibration depends on sample size and class balance; in sparse regimes, rare-but-valid records may overlap with high-penalty regions. Third, the fixed frontier ($H < 0.5$) is practical but not universally optimal; deployment settings may require task-specific threshold calibration and cost-aware tuning, and extending calibration to non-exchangeable data streams via conformal methods [41] remains open. Fourth, the current empirical benchmark set, while diverse, does not exhaust domain types (for example, temporal or relational tables), so external validity should be interpreted cautiously.

5.4 Broader Impact

By providing a verifiable integrity signal, HIF strengthens the case for reproducible research in privacy-sensitive domains. We recommend using HIF primarily as a diagnostic tool to guide generative model refinement rather than as a purely exclusionary filter.

6 Conclusion

As generative models grow in complexity, the metrics used to evaluate synthetic tabular data must evolve beyond aggregate marginal alignment to assess row-level joint distributional consistency. We introduced the Hybrid Integrity Framework (HIF), a probabilistic diagnostic for detecting inter-feature dependency violations in synthetic tabular data, combining the Logical Sentinel Ensemble (LSE) for categorical conditional dependencies with Neighbor-Invariant Continuity (NIC) for continuous feature plausibility within the learned categorical context.

Our empirical results on Census ACS and transaction benchmarks show monotonic HIF degradation with increasing targeted dependency corruption in our experiments, while marginal metrics such as Moment Matching and KS remain static under the same corruption; joint correlation metrics also respond to broad corruption, but only HIF attributes it to specific records. Our cross-architecture audit reveals a persistent "Integrity-Fidelity Decoupling": sophisticated models such as Vine Copulas can achieve high distributional fidelity (KS > 0.96) while failing 85–100% of dependency integrity tests on constrained datasets. By implementing non-compensatory probabilistic aggregation across independently learned dependency models, HIF provides a conservative row-level signal of joint distributional consistency—conformity to the conditional feature dependencies of the real data—that aggregate marginal metrics often miss. We further show that HIF outperforms standard geometric outlier detectors (Isolation Forest, LOF) on conditional dependency violations while remaining appropriately insensitive to marginal noise that does not constitute a dependency failure. Future work will extend this framework to high-frequency financial time-series data [42] and explore kernel-based projections to capture higher-order dependencies [43–45].

Code and Data Availability

The complete open-source implementation, experimental benchmarks, and dataset configurations are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

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Data Availability. The experimental benchmarks and dataset configurations, together with the complete open-source implementation, are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

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A Formal Results for the HIF Score Under Stated Assumptions

We provide a formal analysis of Hybrid Integrity Framework (HIF) score properties under the **structured drift-proof (SDP)** framework introduced by Shang et al. [10]. The results below should be interpreted as assumption-scoped statements rather than unconditional guarantees.

Theorem 1 (Intersection Soundness). The HIF score $H(\mathbf{x})$ defines a membership certificate for the intersection of semantic manifolds $\mathcal{M}_{total} = \bigcap_l \mathcal{M}_l$. Under the idealization that each component validity satisfies $\mathcal{C}_l(\mathbf{x}) = 1$ iff $\mathbf{x} \in \mathcal{M}_l$ and $\mathcal{C}_l(\mathbf{x}) \in [0, 1)$ otherwise, $H(\mathbf{x}) = \left(\prod_l \mathcal{C}_l(\mathbf{x})\right)^{1/L}$ attains its maximum value 1 exactly on $\mathcal{M}_{total}$:

$$\nVdash[H(\mathbf{x}) = 1] = \prod_l \nVdash[\mathbf{x} \in \mathcal{M}_l] \tag{10}$$

Proof. *Completeness:* if $\mathbf{x} \in \mathcal{M}_{total}$ then $\mathcal{C}_l(\mathbf{x}) = 1$ for all l, so $H(\mathbf{x}) = 1$. *Soundness (non-compensatory property):* if $\mathbf{x} \notin \mathcal{M}_{total}$, there exists l^* with $\mathcal{C}_{l^*}(\mathbf{x}) < 1$, and since all validities lie in $[0, 1]$, the geometric mean satisfies $H(\mathbf{x}) < 1$. *Strict convergence:* an "Absolute Violation" in any layer ($\mathcal{P}_{l^*} \to 1$) drives $H(\mathbf{x}) \to 0$ regardless of other layers. $H(\mathbf{x})$ thus behaves as a conservative diagnostic for joint satisfaction of latent manifold constraints, matching the implementation in Algorithm 2.

Theorem 2 (Asymptotic Convergence of the HIF Penalty). Let $\hat{P}_N(x_h \mid \mathbf{x}_{\setminus h})$ be the empirical conditional probability estimator (the Sentinel f_h) trained on a ground-truth dataset $\mathcal{D}_{real}$ of size N. Assuming the base estimator is statistically consistent (e.g., consistent non-parametric trees), as $N \to \infty$, $\hat{P}_N$ converges in probability to the true conditional distribution $P^*(x_h \mid \mathbf{x}_{\setminus h})$, and the empirical penalty $\mathcal{P}_h^{(N)}(\mathbf{x})$ converges pointwise in probability to the corresponding population penalty:

$$\lim_{N\to\infty} P\left(\left|\mathcal{P}_h^{(N)}(\mathbf{x}) - \mathcal{P}_h^*(\mathbf{x})\right| > \epsilon\right) = 0 \quad \forall \epsilon > 0,\ \mathbf{x} \in \mathcal{M}_{real} \tag{11}$$

Proof sketch. *(i)* Because the LSE leverages consistent non-parametric estimators, the empirical conditional probabilities converge pointwise in probability to the true Bayesian posterior P^* (provided the feature space is bounded and leaf nodes grow sub-linearly with N); the pointwise statement reflects that sup-norm convergence is not generally guaranteed for forest-based estimators. *(ii)* Defining the confidence floor δ_h as the empirical 5th-percentile of the sentinel's out-of-bag probabilities yields an asymptotic upper bound of approximately $\alpha = 0.05$ on false positives under calibration assumptions. *(iii)* Records penalized by the converged HIF ($\mathcal{P}^* > 0$) are expected to lie in lower-probability conditional regions of the manifold, supporting a structural-violation interpretation. *(iv)* By the Uniform Law of Large Numbers and the Continuous Mapping Theorem, the empirical Feature Dependency Score $S^{(N)}(h)$ converges in probability to the Bayes optimal classification accuracy $S^*(h)$, implying asymptotically stable hub ranking.

Theorem 3 (Type I Error Bound & Cohort-Purity Approximation). Let $\mathbf{x} \sim P_{real}$ be a valid record on the ground-truth manifold $\mathcal{M}_{real}$ (including rare tail outliers). Under quantile calibration ($\delta_h = Q_\alpha$ and continuous tolerance bandwidth $\gamma_y = Q_{1-\beta}$), the false-rejection probability is upper-bounded by the Union Bound:

$$P\left(H(\mathbf{x}) < 0.5 \mid \mathbf{x} \in \mathcal{M}_{real}\right) \leq \sum_{h\in\mathcal{H}} \alpha_h + \sum_{y} \beta_y \tag{12}$$

Furthermore, let a generative model produce a noisy mixture $P_{syn} = (1-\eta)P_{real} + \eta P_{hallucinated}$, where $\eta \in [0, 1]$ is the structural hallucination rate. Rejection filtering with threshold $H \geq 0.5$ yields an approximate retained-cohort purity:

$$\text{Purity}(H \geq 0.5) = \frac{(1-\eta)(1-\alpha)}{(1-\eta)(1-\alpha) + \eta\beta_{hallucination}} \tag{13}$$

where $\beta_{hallucination}$ is the leakage probability of hallucinated records past the filter. For high hallucination rates ($\eta \approx 0.66$), the pre-filter purity is $(1-\eta) = 34\%$; with a calibrated leakage rate $\beta_{hallucination} \approx 0.02$ and $\alpha = 0.05$, filtering elevates purity to over 96% in this illustrative model, theoretically accounting for the downstream predictive utility improvements observed.

B End-to-End Auditing Procedure

Algorithm 1 summarizes the complete manifold calibration procedure—hub selection, confidence-floor calibration, NIC regressor training, and rule mining—and Algorithm 2 details the end-to-end auditing procedure for scoring a synthetic dataset with the trained HIF components.

Algorithm 1 HIF Manifold Calibration

Input: Ground-truth data $\mathcal{D}_{real}$, Hub count K_{hub}
Output: Sentinels $\{f_h\}$, Floors $\{\delta_h\}$, NIC Regressors $\{r_y\}$, Thresholds $\{\tau_y\}$

$\mathcal{H} \leftarrow$ Select top K_{hub} features by Feature Dependency Score $S(h)$
for each hub $h \in \mathcal{H}$ **do**
    Train Sentinel $f_h : \mathcal{X}_{\setminus h} \to [0,1]^{|\mathcal{X}_h|}$ on $\mathcal{D}_{real}$ to predict x_h
    $\delta_h \leftarrow \max(\text{Percentile}_5(\{f_h^{\text{OOB}}(\mathbf{x}_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}), 0.01)$
end for
$\mathcal{Z} \leftarrow \text{SVD}(\text{Scale}(\text{OneHot}(\mathcal{D}_{cat})))$ ▷ Non-Centered Spectral Manifold
for each continuous feature y **do**
    Train Gradient Boosting regressor $r_y : \mathcal{Z} \to \mathbb{R}$ on $\mathcal{D}_{real}$
    $\mathcal{E}_y \leftarrow \{|y - r_y(\mathbf{z})| \mid \mathbf{x} \in \mathcal{D}_{real}\}$ ▷ Ground-truth residuals
    $\tau_y \leftarrow \max(Q_{0.95}(\mathcal{E}_y), \text{Median}(\mathcal{E}_y) + 2 \cdot \text{MAD}(\mathcal{E}_y))$ ▷ Robust Threshold
    $\gamma_y \leftarrow \max(Q_{0.98}(\mathcal{E}_y), 2 \cdot \tau_y, 3 \cdot \text{MAD}(\mathcal{E}_y), 0.1)$ ▷ Dynamic Gamma Scaling
end for
$\mathcal{R} \leftarrow \text{MineImplicationRules}(\mathcal{D}_{real})$ ▷ Apriori-style, min_conf=0.95, min_supp=0.005

C Logical Sentinel Ensemble (LSE) Implementation Details

On the U.S. Census ACS dataset, the Feature Dependency Score $S(h)$ identified the following top-5 hubs: `household_income`, `housing_cost`, `poverty_status`, `education`, and `tenure`. The confidence floor δ_h is the 5th-percentile of the *out-of-bag* probability mass that a sentinel assigns to the correct category in the ground-truth data (safety floor 0.01); out-of-bag predictions yield out-of-sample floors without a held-out split, reflecting generalization rather than memorized fit, and avoid the instability of 1st-percentile thresholds in sparse data [6, 46].

Algorithm 2 HIF Synthetic Data Audit

Input: Synthetic record $\mathbf{x}$, Trained components from Alg 1
Output: Record Integrity Score $H(\mathbf{x})$

Step 1: Logical Sentinel Audit (Categorical)
for each hub $h \in \mathcal{H}$ **do**
  $\mathcal{P}_h \leftarrow \text{clip}\left(\frac{\delta_h - f_h(\mathbf{x}_h|\mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right)$
end for
$\mathcal{P}_{cat} \leftarrow 1 - \prod_{h\in\mathcal{H}}(1 - \mathcal{P}_h)$ ▷ Multiplicative violation aggregation
Step 2: Neighbor-Invariant Continuity Audit (Continuous)
$\mathbf{z} \leftarrow \phi(\mathbf{x}_{cat})$ ▷ Apply learned standardized projection
for each continuous feature y **do**
  $\rho_y \leftarrow |y - r_y(\mathbf{z})|$
  $\mathcal{P}_y \leftarrow \text{clip}\left(\frac{\rho_y - \tau_y}{\gamma_y}, 0, 1\right)$
end for
$\mathcal{P}_{cont} \leftarrow \max_y \mathcal{P}_y$
Step 3: Integrity Aggregation
$\mathcal{P}_{rule} \leftarrow 1$ if $\mathbf{x}$ violates any $r \in \mathcal{R}$ else 0
$H(\mathbf{x}) \leftarrow \left(\prod_l (1 - \mathcal{P}_l)\right)^{1/L} \in [0, 1]$ ▷ Geometric mean of validities

D Extended Results and Sensitivity Analysis

D.1 Hyperparameter Sensitivity

We evaluated HIF robustness across its three key hyperparameters on the Census ACS dataset (2,000 records with 5% injected violations; $N = 5$ seeds, mean ± SEM). The continuous HIF score is the primary metric; the violation rate derives from applying the violation threshold to row-level penalties. The HIF score is stable across hub counts (varying by $< 1\%$ from 0.979 ± 0.0003 at $K = 1$ to 0.987 ± 0.0007 at $K \geq 10$), since geometric-mean aggregation prevents added sentinels from diluting the score; the hub set saturates at $K = 10$ because Census ACS contains only 10 features. Scores are likewise stable for confidence-floor percentiles 1–5 (0.9848 ± 0.0002 at percentile 1 vs. 0.9841 ± 0.0002 at percentile 5; minor 0.3% drop to 0.9815 ± 0.0002 at the 10th), making the 5th-percentile default a good sensitivity–robustness balance. Finally, the continuous score is invariant to the binary threshold (0.9841 ± 0.0002 for thresholds 0.3, 0.5, 0.7), while the violation rate scales monotonically (0.81% → 0.37% → 0.21%), letting practitioners tune the operating point without retraining.

D.2 Scalability and Baseline Comparisons

We benchmarked HIF against the **SHAP Distance** semantic fidelity metric [9], which serves as a state-of-the-art baseline for explainability-aware evaluation. Although the SHAP Distance effectively identifies semantic drift, it suffers from severe scalability bottlenecks. HIF auditing scales linearly, $O(N \cdot K)$, where K is the number of hubs. In our benchmark (`scripts/11_scalability_benchmark.py`), fitting the 5-hub LSE auditor on 2,000 real records took 9.7 s, after which scoring 10,000 synthetic records took 1.50 ± 0.15 s and scoring 100,000 records took 10.6 ± 0.5 s on commodity multi-core hardware (fit amortized once; wall-clock averaged over three repeats). The marginal per-10,000-row cost at the 100,000-row scale ($10.6 / 10 \approx 1.1$ s per 10k rows) confirms the linear scaling. In contrast, SHAP-based methods require $O(N \cdot D \cdot 2^D)$ operations or expensive approximations, which precludes exact evaluation at this volume. Thus, HIF provides comparable detection accuracy while maintaining the computational efficiency required to scale to high-cardinality datasets with millions of rows.